Python Starter Guide for Beginners

Eight small ideas that get you writing real programs. Read the concept, run the example, compare with the sample output.

What you will learn

1. Your first program
2. Variables
3. Data types
4. Input from the user
5. Operators
6. Decisions: if / else
7. Loops
8. Lists
9. Functions

1. Your first program

print() shows text on the screen.

Explanation

Python runs your code line by line, top to bottom. Text goes inside quotes. Install Python from python.org, save your code in a file ending in `.py`, and run it with `python hello.py`.

Example

```
print("Hello, Entri!")
print("I am learning Python")
```

Sample output

```
Hello, Entri!
I am learning Python
```

2. Variables

A variable is a named box that stores a value.

Explanation

Use `=` to put a value in the box. Names can use letters, numbers and underscores, cannot start with a number, and are case-sensitive. Use `#` to write a comment that Python ignores.

Example

```
# store values
name = "Naren"
age = 25
print(name)
print(age)
```

Sample output

```
Naren
25
```

3. Data types

Every value has a type: str, int, float or bool.

Explanation

- `str` : text, like "Python"
- `int` : whole numbers, like 10
- `float` : decimal numbers, like 9.5
- `bool` : True or False

Use `type()` to check. Use an f-string (`f"..."`) to place variables inside text.

Example

```
course = "Python"
```

```
fee = 4999.50
is_open = True
print(type(course), type(fee), type(is_open))
print(f"{course} course fee is {fee}")
```

Sample output

```
<class 'str'> <class 'float'> <class 'bool'>
Python course fee is 4999.5
```

4. Input from the user

input() reads what the user types, always as text.

Explanation

To do maths with the answer, convert it first: `int()` for whole numbers, `float()` for decimals.

Example

```
name = input("Enter your name: ")
year = int(input("Enter your birth year: "))
print(f"Hi {name}, you are about {2026 - year} years old")
```

Sample output

```
Enter your name: Priya
Enter your birth year: 2000
Hi Priya, you are about 26 years old
```

5. Operators

Operators calculate and compare values.

Explanation

- Maths: `+ - * /` , plus `//` (whole-number division), `%` (remainder), `**` (power)
- Comparison: `== != > < >= <=` , giving True or False

- Logic: `and` , `or` , `not`

Example

```
a = 17
b = 5
print(a + b, a / b, a // b, a % b, a ** 2)
print(a > b and b > 10)
```

Sample output

```
22 3.4 3 2 289
False
```

6. Decisions: if / else

Run different code depending on a condition.

Explanation

End the condition line with a colon and indent the lines below it (4 spaces). Python uses indentation to know which lines belong together. Use `elif` for extra choices.

Example

```
marks = 72
if marks >= 90:
    print("Grade A")
elif marks >= 60:
    print("Grade B")
else:
    print("Keep practising")
```

Sample output

```
Grade B
```

7. Loops

Repeat code without copying it.

Explanation

`for` repeats a set number of times. `range(1, 6)` gives 1 to 5 (the end number is not included). `while` repeats as long as a condition stays True.

Example

```
for i in range(1, 6):
    print("Day", i)

count = 3
while count > 0:
    print(count)
    count = count - 1
print("Go!")
```

Sample output

```
Day 1
Day 2
Day 3
Day 4
Day 5
3
2
1
Go!
```

8. Lists

A list stores many values in one variable.

Explanation

Items sit inside square brackets. Counting starts at 0, so the first item is `[0]` and the last is `[-1]`. Use `append()` to add, `len()` to count, and a `for` loop to visit each item.

Example

```
skills = ["Python", "SQL", "Excel"]
skills.append("Power BI")
print(skills[0])
print(len(skills))
for skill in skills:
    print("-", skill)
```

Sample output

```
Python
4
- Python
- SQL
- Excel
- Power BI
```

9. Functions

A function is reusable code with a name.

Explanation

Define it with `def` . Values you pass in are called parameters. `return` sends a result back. Write it once, call it as many times as you need.

Example

```
def greet(name):
    return f"Welcome, {name}!"

def add(x, y):
    return x + y

print(greet("Arun"))
print(add(7, 8))
```

Sample output

```
Welcome, Arun!
15
```

Next steps

Type every example yourself instead of copying, then change the values and watch the output change. Once these nine ideas feel comfortable, move on to dictionaries, file handling and small projects.

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